

DO CONCENTRATED TRADING EQUILIBRIA EXIST?

THE MIGRATION OF INFORMED TRADING FOLLOWING INDEX ADDITION

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Abstract

In this paper, we use *index addition* to gain insights into the trading behavior of informed traders. When a stock is added to the S&P 500 index, it attracts permanent interest from index funds, which are, by definition, liquidity traders. Consequently, this event is associated with an exogenous increase in the volume of liquidity trading, which is likely to be concentrated at the end of the day and in large-sized trades. These aspects of fund trading, in turn, change the optimal strategies of informed traders. We can characterize the resulting equilibrium by comparing ‘steady state’ trading patterns before and after index addition. Our key findings are as follows. *First*, while both daily and intraday volume increase substantially after index addition, the largest increase occurs in the last half-hour of the day. There is a permanent increase in daily and intraday return volatility after addition; once again, the largest increase occurs at the end of the day. The increase in return volatility throughout the day suggests that increased information production follows index addition. However, the larger increase in end-of-day volatility also suggests that informed traders change the timing of their trades to coincide with liquidity trades. *Second*, the distribution of trade sizes shifts rightwards after addition. However, the price impact of individual large trades does not increase. This creates incentives for informed traders to make large trades, since they are now able to ‘hide’ these trades more easily. Consequently, the proportion of total end-of-day price change occurring on large trades increases significantly after addition. Our results support the conclusion that informed traders alter their trading strategies to optimally match both the timing, and the size, of liquidity trades in the market.

* The authors thank Tim Poole for research assistance, and seminar participants at the University of Alberta and the University of Oregon for helpful comments. The second author acknowledges financial support from a Pearson fellowship at the University of Alberta’s Faculty of Business.

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Introduction

The interaction between informed and liquidity, or uninformed, traders is central to much of market microstructure theory. In the many theoretical models that have been developed, the basic setting almost always involves liquidity traders and informed traders.¹ These models predict that the optimal strategies of informed traders regarding the timing and size of their trades will depend on the corresponding patterns in liquidity trade. It is difficult to evaluate these predictions since the timing and size of liquidity trades are unobservable. In this paper, we use 246 additions to the Standard and Poor's 500 index between 1983 and 1996 to explore some of the predictions related to the strategies of informed traders.

When a stock is added to the S&P 500, it attracts enduring interest from index funds. These funds trade the stock, not only in the immediate aftermath of the addition, but also on a recurring basis, as they experience cash inflows or outflows. Cash flows into index funds have averaged 10% of mutual fund flows in recent years.² Thus, the trading activity of index funds is substantial. The trades of index funds are motivated, not by private information about the stocks in the index, but by the objective of minimizing tracking error relative to the S&P 500. Hence, index funds can be classified as liquidity traders. In

¹ O'Hara [1995] provides an excellent introduction to theoretical and empirical work in market microstructure.

² Estimates reported in the *Wall Street Journal* [January 24, 1997].

addition, index funds have strong incentives to trade near the end of the day (when the value of the S&P index is computed) and are likely to make large-sized trades. We argue, therefore, that index addition is associated with a *substantial and exogenous* increase in the volume of liquidity trade, which is likely to be concentrated at the end of the day and in large trades.³

We exploit this change in the trading environment to examine two issues related to the notion of a concentrated trading equilibrium. The first issue relates to the temporal concentration of trading. Specifically, we examine whether, driven by their desire to trade when the market is “thick”, informed traders attempt to trade when liquidity traders are active (Admati and Pfleiderer [1988]). Since informed trades provide information to the market, the observable implication of the temporal concentration equilibrium is a permanent increase in return volatility at the end of the day. The second issue relates to concentration in trades sizes. Informed traders attempt to choose the size of their trades to match the distribution of liquidity trades in the market (Kyle [1985], Barclay and Warner [1993]). Since the frequency of large liquidity trades increases after addition, informed traders are able to submit large trades. The implication of this hypothesis is that the proportion of information revealed through large trades increases after addition.

³ The addition of a stock to the S&P 500 also generates trading interest from index arbitrageurs who attempt to profit from temporary deviations between the S&P futures and cash indexes through appropriate transactions in the futures and in the underlying stocks. However, while trades related to index arbitrage are likely to be large, they should not be concentrated at the end of the day. Hence, while program trades may be a factor in the post-addition trading environment, we feel that they are unlikely to explain our results at the end of the day. For evidence, see Hasbrouck [1996] who documents that 85% of index-arbitrage trades are market orders and that market-on-close orders occur mainly on days when index futures and futures option contracts expire.

Our investigation of these issues proceeds by comparing trading activity for the added stock over the year preceding, and the year succeeding, index addition. We exclude the period immediately surrounding the addition since our objective is to examine the ‘steady state’ trading environment.

We find that the trading environment changes dramatically following index addition. Trading volume and the number of transactions increase significantly, and the largest increase in each occurs at the end of the day. Additionally, the distribution of trade sizes shifts rightward, and this shift is most marked at the end of the day. Each of these findings is consistent with index funds being present in the market at the end of the day. The bid-ask spread and the price impact of large trades decline throughout the day. Thus, it appears as if the market maker recognizes, and responds to, the increased proportion of liquidity traders in the market.

However, the decline at the end of the day in the spread and the price impact is similar to the corresponding reduction at other times, suggesting that the market-maker does not perceive a greater alleviation of the adverse selection problem at the end of the day. In other words, the increased frequency of large liquidity trades at the end of the day creates incentives for more informed traders to place large trades at this time as well. We find that daily and intraday return volatility increase after addition, with the largest increase occurring at the end of the day. These findings do not appear to be driven by price pressure as the autocorrelation in returns does not become more negative after addition. Rather, the permanent across-the-board increase in volatility is consistent with increased information production following index addition. The pronounced increase in volatility at

the end of the day is consistent with informed traders altering the timing of their trades to coincide with liquidity trades.

Following Barclay and Warner (1993), we assess the informativeness of trades by calculating the proportion of the total price change for a given period occurring in various trade size categories, and find that the informativeness of large trades increases at the end of the day. Overall, our results suggest that informed traders migrate to the end of the day and to larger trade sizes when the new trading environment established by the activity of index funds allows them to better mask their trades. In other words, our results suggest that informed traders alter their strategies to optimally match both the timing, and the size, of liquidity trades in the market, creating a concentrated trading equilibrium.

The rest of this paper is organized as follows. In Section 1 we explain why index addition is an interesting event to examine microstructure questions. Section 2 discusses the questions that we aim to address in this study. Section 3 describes the data and Section 4 the results. Section 5 discusses the implications of these results and concludes.

1 Why is looking at index addition interesting from a microstructure perspective?

Numerous indexes are followed or ‘tracked’ by mutual funds. Of these, the Standard and Poor’s 500 (S&P 500) is the most prominent and popular. When a stock is added to the S&P 500 index, such *index funds* (*funds* for short) buy the stock in order to rebalance their portfolios and track the revised index.⁴ Subsequently, flows into or out of funds

⁴ The discussion that follows is couched in terms of the S&P 500 index since this index is the focus of our paper. The same issues are obviously relevant in the case of addition to any index (e.g. the Russell 2000).

necessitate additional trading on their part. Thus, index addition leads to an enduring increase in fund trading of the stock.⁵ Consistent with this fact, Dhillon and Johnson [1991] and Beneish and Whaley [1996] document a permanent increase in trading volume after a stock is added to the S&P 500 index.

Index funds do not profess to possess private information when they buy or sell stocks in the index that they track. In fact, the avowed objective of index funds is to passively track the index, and to minimize both tracking error and transactions costs. As an example, we provide the following descriptions, drawn from the Vanguard Index Trust prospectus, of the objectives and investment strategies of the Vanguard S&P index fund.

(a) “..(I)f 5% of the S&P 500 Index were made up of the assets of a specific company, the 500 Portfolio would invest the same percentage of *its* assets in that company.”

(b) “Generally, .. [the fund] sells securities only to respond to redemption requests or to adjust the number of shares held to reflect a change in the .. index.”

The 500 Portfolio holds 500 stocks and reports that the fund return is within 20 basis points of the S&P 500 Index return (the correlation between the fund and index returns is 1.0). On the basis of the stated objectives of index funds, it is appropriate to regard index funds as uninformed, or liquidity, traders and, accordingly, to interpret the trading volume they generate as liquidity trade.

Furthermore, index funds have a clear preference for trading near the close. First, they are able to minimize tracking error in their returns relative to their benchmark (the S&P

⁵ Stocks added to the S&P 500 index experience permanent positive abnormal returns. These abnormal returns are attributed to a rightward shift in demand. See Shleifer [1986] and, for a dissenting view, Harris and Gurel [1986] and Dhillon and Johnson [1991], for instance.

500), whose value is computed using closing prices. Also, by waiting and thereby being able to offset individual inflows and outflows occurring during the day, index funds can reduce the required amount of trading and their trading costs. Both arguments suggest that index funds will be active near the end of the trading day.⁶ Realistically, however, funds are able to precisely compute their net inflows for the day only after the market closes. Hence, there is the possibility of spillover effects after the following open as funds deploy the residual flows.

Finally, as institutional traders in a deep market, it is unlikely that index funds frequently make small trades, say, a single round lot. In other words, index funds are expected to make large trades on a regular basis (we define ‘large’ trades later).

Putting these three features of index fund trading together, we argue that index addition is accompanied by a substantial increase in liquidity trading activity. Furthermore, the increased liquidity trading is likely to be concentrated at the end of the day. Finally, the majority of this increased trading occurs in large-sized trades. In short, there is an exogenous rightward shift in the distribution of liquidity trades, and this shift is concentrated at the end of the day.

A key premise in this paper is that index fund trading of the added stock is an important part of overall trading activity. In order to assess the validity of our assumption, we examine trading volume and the distribution of trade sizes before and after

⁶ Anecdotal evidence supports these arguments. In describing the daily activities of the largest index fund, the Vanguard Index Trust 500 Portfolio, the *Wall Street Journal* [January 28, 1997] reports that the fund “has been buying some stocks all day” but that “the bulk of its buying .. will occur in the next 45 minutes, right before the close of the New York Stock Exchange at 4:00 p.m. and the Chicago Mercantile Exchange ..[at] 4:15 p.m.” In the same newspaper [March 3, 1997] the fund’s manager describes why

addition to search for tangible evidence that index funds are active. We find that, after addition, volume increases, and the size distribution of trades shifts rightwards and that the shift in each is most pronounced at the end of the day (these statistics are presented in greater detail below).

However, we also look for published statistics that shed light on the importance of index funds. According to Morningstar estimates reported in the *Wall Street Journal* [May 26, 1998], \$600 billion (10% of mutual fund assets) is currently invested in index funds. The proportion of aggregate mutual fund inflows directed at index funds has been fairly steady in recent years, at approximately 10 percent.⁷ Informal indexing strategies are adopted in-house by the managers of corporate pension plans. It is difficult to get concrete estimates of aggregate inflows corresponding to such plans, but it is conceivable that these are of similar magnitude to inflows for index funds.⁸ Taken together, these statistics suggest that trading on the part of institutional investors tracking the S&P 500 index is an important component of normal trading activity.

Some caveats

Several factors reduce the power of our tests. The visibility of closing prices makes it attractive for investors other than index funds to trade at the end of the day. Similar motivations (to index funds) may cause other mutual funds to trade near the end of the day. Brokers may choose to wait until the end of the day to trade on behalf of their

one-third of his trades occur in the final hour: “[S]ince the investor is buying the fund at the closing price of the day, we want to pay a price as close as possible to that on close.”

⁷ See, for instance, the *Wall Street Journal* [January 24, 1997 and January 28, 1997].

⁸ For example, see the *Wall Street Journal* [December 23, 1996].

clients. To the extent that such trades are widespread before and after index addition, the microstructure effects associated with index fund trading may be difficult to identify.

Index funds may rationally choose to trade earlier in the day, incurring a larger tracking error but saving on trading costs (since they are able to avoid the end-of-day stampede.) These trading costs include the bid-ask spread and the price impact, if any, of the transactions. Additionally, many funds probably follow fewer than 500 stocks. For our purposes, however, as long as most funds do not ignore the same stocks, the effects of fund trading on every added stock should be detectable. Finally, funds can trade in the S&P futures contract rather than in the underlying stocks. This strategy is a useful way of investing funds flexibly and at low cost, but will lower the required amount of trading in the stock market. Each of these aspects of fund trading will also reduce the likelihood of detecting microstructure effects in trading activity for the added stock.

2 Testable hypotheses

Section 1 has argued that adding a stock to the S&P 500 index results in an exogenous increase in the volume of liquidity trading. Interesting predictions in market microstructure theory relate to the effects of liquidity trading on strategic decisions by informed traders regarding the timing and size of their trades. In trying to test these predictions, researchers confront two related problems. First, the volume of liquidity trading is unobservable, so it is impossible to empirically separate volume into liquidity and informed components. Second, from a theoretical perspective, the volume of both liquidity and informed trading is jointly determined as the outcome of optimization decisions on the part of liquidity and informed traders.

In order to see the advantages of our experiment, it is useful to think of two equilibria for the added stock, one reflected in the pre-addition trading environment, and the other in the post-addition environment. The difference between the two environments is associated with effects related to the trading activity of index funds. In other words, the post-addition trading environment reflects an exogenous shift in the volume of liquidity trading and the reaction (if any) of informed traders. Examining the effects of index addition on the trading environment, therefore, constitutes a powerful and unique way of studying equilibrium predictions related to liquidity trade.

We consider two questions related to the notion of a concentrated equilibrium. The first relates to temporal concentration of trading, i.e. whether periods of heavy liquidity trading are also characterized by heavy informed trading and, therefore, by high volatility. The second deals with concentration in trade sizes, i.e. whether an increase in the typical trade size for liquidity traders results in an increase in the typical size of an informed trade, which, in turn, affects the informativeness of trades of different sizes. A discussion of these issues follows.

Do patterns in liquidity trading lead to concentration of trading in time or trade size?

The increase in liquidity trade resulting from index addition alters the incentives of informed traders to collect private information. All else being equal, an increase in liquidity trading reduces the adverse selection problem faced by the market maker, and, therefore, reduces trading costs (the spread and/or depth). This translates into higher

returns to information production, and leads to more information being produced (Admati and Pfleiderer [1988]).

The changed patterns in liquidity trade also affect the optimal choice by informed traders of the time and size of their trades. Informed investors attempt to trade strategically in order to maximize the value of their private information.

Concentration in Time

Informed traders have strong incentives to trade when the market is “thick”, i.e. when liquidity traders are active. (Admati and Pfleiderer [1988]).⁹ Return volatility increases at this time as well, as the formerly private information is revealed.

If index fund trading results in a sharp increase in liquidity trading at the end of the day, some informed traders should migrate to the end of the day as well. Under the *concentrated temporal equilibrium hypothesis*, therefore, return volatility should increase at the end of the trading day. If information production is unchanged by index addition, this increased volatility will come at the expense of volatility earlier in the day (as informed traders shift some of their trades to the end of the day). However, if index addition is accompanied by an increase in the production of private information, and by an increase in liquidity trading throughout the day, informed trading and, therefore, return volatility will increase throughout the day. Nevertheless, if the largest increase in liquidity trading occurs at the end of the day, the sharpest increase in volatility will also occur at this time.

⁹ This is an equilibrium because the (risk-neutral) informed traders compete with each other and thereby reduce the cost of trading to the liquidity traders. Subrahmanyam [1991] shows that the Admati-Pfleiderer conclusions are sensitive to the assumption of risk-neutrality.

We first verify that the largest increase in trading volume occurs at the end of the trading day. This confirms the increased presence of index funds in the stock market at the end of the day and provides a new focal point for informed traders. We then examine the *concentrated temporal equilibrium* prediction that the most pronounced increase in volatility will occur at the end of the day.

Concentration in Trade Size

We also study a second aspect of concentration. This is motivated by the literature dealing with the choice of trade sizes by informed traders. Kyle [1985] and Barclay and Warner [1993] argue that an informed trader would like to acquire a large position but is forced to strategically break up his trades in order to avoid tipping the market maker off about his information. Consistent with this ‘stealth trading’ hypothesis, Barclay and Warner show that most of the cumulative price change occurs on medium-sized transactions (500-9900 shares) despite their low share of overall trading volume. In general, however, the informed trader’s optimal trade size should depend on the existing distribution of liquidity trades.

We have argued above that index funds will choose to trade in large-sized lots. As a result, there is an increase in the volume of liquidity trading occurring in large-sized trades. The stealth-trading hypothesis then predicts that informed traders have incentives to place large trades. Formerly, these trades would have tipped the market maker off and brought about a significant price revision. Now, however, the market maker reacts less to these trades because he knows that a larger proportion (than previously) consists of non-informational trades. Since the trades of the informed reveal their information, the

proportion of the total price change that occurs on large trades will increase. The *concentrated trade size equilibrium hypothesis*, therefore, suggests the presence of a concentrated equilibrium in trade sizes, with the gravitation to large trades after index addition.

If index funds trade at the end of the day and the trades of funds tend to be large, then after addition, there should be a rightward shift in the size distribution of trades at the end of the day. Informed traders can exploit this increase in the frequency of large liquidity trades by increasing the number of large trades they make at the end of the day. Thus, the increase in the informativeness of large trades should be most striking at the end of the day.

An alternative explanation for the elevated volatility at the end of the day is based on ‘noisy prices’ (see, for instance, French and Roll [1986] and Black [1986]), or price pressure. The hypothesis is that increased trading by index funds generates temporary mispricing, which also results in higher volatility. Under this *price pressure* hypothesis, the mispricing is corrected and returns are negatively autocorrelated. By contrast, if the high volatility results from the revelation of additional private information, it should not be reversed. We will directly test the prediction of the price pressure hypothesis as we present our results.

Relation to other empirical studies

A few papers explore the effects of other events on liquidity trading. Jegadeesh and Subrahmanyam [1993] study the effects of the introduction of the S&P futures contract on the bid-ask spread for stocks that were part of the S&P 500 Index at the time. Mayhew

et. al [1995] consider the effects of changes in option market margin requirements on liquidity and the price impact of trades in both the option market and the market for the underlying stock. Kumar et. al [1998] examine the effects of option introduction on spreads, depth, volume, and trade size in the market for the underlying. Each of these papers examines the effects of an event on the general characteristics of trading.

We carry out a similar analysis, but we also examine richer predictions. Our most interesting empirical test deals with trade sizes. Since trading by index funds is likely to be concentrated in large-sized trades, we can examine whether there is a shift in the strategies of the informed traders. Second, we examine the effects on intertemporal patterns, studying the possible gravitation to the end of the day, as predicted by the Admati and Pfleiderer [1988] model.¹⁰

Beneish and Whaley [1996] analyze the effects of index addition over 1986-1994, and provide some statistics that are similar to our daily statistics. They find that volume increases permanently, and that there is a temporary increase in trade sizes and a temporary reduction in the bid-ask spread. Their focus, however, is on front running and its effects on trading activity.¹¹ Consequently, Beneish and Whaley deliberately take a short-term view of the event, while our objective is to examine the normal trading patterns that result from a permanent shift in the trading environment.

¹⁰ Barclay, Litzenberger and Warner [1990] analyze the effects of increased trading hours, associated with Saturday trading on the Tokyo Stock Exchange, and find unchanged weekly variance but higher weekend variance in weeks with Saturday trading. Our analysis of intraday and daily concentration is similar to theirs in that we examine the effects on variance of increased trading.

¹¹ Starting in October 1989, Standard and Poor's has announced changes in the composition of the index one week before the changes become effective. This creates incentives for front-running, or purchasing ahead of funds, which tend to buy on the effective date.

3 Data and Empirical Methods

We examine a sample of 246 index additions during the period from 1983 to 1996. Our choice of this interval is driven by the availability of intraday data. The list and dates of index changes are obtained from the Standard and Poor's [1996] *Security Price Record*.

We wish to examine 'steady state' market activity on either side of index addition. Consequently, data on intraday prices, volume and number of trades at half-hourly frequencies are collected for a two-year period centered on the date the stock is added to the S&P 500 index. The data come from the Institute for the Study of Securities Markets (ISSM) tapes for 1983-92 and from Trade and Quote (TAQ) CDs for 1993-96. We compare market statistics over the *pre-addition* period, consisting of trading days $(-260, -20)$ relative to the event, and the *post-addition* period, consisting of trading days $(+20, +260)$. We ignore the two months immediately surrounding the announcement because this is a transitional period and many of the days in this period are unlikely to be normal. Additionally, potentially confounding effects are present. For instance, Beneish and Whaley [1996] document the upsurge in arbitrageur trading activity around the addition date. Furthermore, the large volume of trading required of index funds might cause them to trade in the middle of the day.

The concentrated equilibrium hypotheses relate specifically to the end of the day. As a result, much of our analysis focuses on intraday patterns. Funds may, however, choose to trade at other times in the day. Hence, we also analyze daily patterns before and after addition. Prior analysis of intraday patterns reveals that the middle of the trading day is characterized by a lower level of trading activity than either the beginning or the end of the

day (see for instance, Wood et. al [1985], McNish and Wood [1992], and Foster and Vishwanathan [1993]). In the intraday analysis, therefore, we divide the day into 5 periods: 9:30-10:00 a.m. (period 1), 10:00-10:30 a.m. (period 2), 10:30 a.m.-3:00 p.m. (period 3), 3:00-3:30 p.m. (period 4), 3:30-4:00 p.m. (period 5). Periods 1 and 2 correspond to the beginning of the day, periods 4 and 5 to the end of the day, and period 3 to the middle of the day. The effects of index addition on the trading environment should be most pronounced in period 5 and perhaps in periods 4 and 1 as well. Before September 30, 1985, the market did not open until 10 a.m. For data from before September 30, 1985, the first two periods were shifted appropriately and the lost half-hour was absorbed by the period 3 (making it 11:00 a.m. to 3:00 p.m.). To make the raw statistics (volume, turnover and volatility) for period 3 comparable to those for the other periods, we scale the statistics for period 3 by the number of half-hours it represents (8 before September 30, 1985 and 9 since that date).

4 Results

In this section we present the results of the empirical tests of the concentrated equilibrium hypotheses. We start by examining the overall trading environment—volume, number of trades, average trade size and volatility. We then study the price impact of trades of different sizes before and after the addition. Finally, we look for evidence of a new concentrated trading equilibrium at the end of the day.

The Trading Environment: Volume, Number of Trades, Trade Sizes and Volatility

Table 1 presents several comparisons of key characteristics of the trading environment in the pre-addition and post-addition periods. The table starts with grand medians for the

52,795 stock-days before and the 58,504 stock-days after the addition.¹² The median daily volume rises by 37% from 95,900 to 131,700 shares. The well-established U-shaped pattern in intraday volume (e.g. Foster and Vishwanathan [1993]) is evident in our sample as well. In fact, the intraday increases in volume indicate that volume becomes more U-shaped post-addition. The middle of the day sees an increase in median volume of 39%, but volume in the last half-hour increases by 71%. While the average half-hour in the middle of the day accounts for approximately 6% of the overall increase in daily volume, the half-hours at the beginning and end of the day each account for over 11% of the increase in daily volume.

As a second, and less noisy, measure of the change in volume from pre to post-addition, we calculate paired volume ratios. For each stock, the mean volume for each day and each intra-day period is calculated for the pre and post-addition periods. Then, the ratio of these mean volumes is calculated, by stock. We present the median of the distribution of these 246 ratios for each intra-day period and for the whole day. The ratios range from 1.23 to 1.43 and every ratio is significantly greater than one, confirming the overall large increase in volume after addition. Once again, the largest increase comes in the last half-hour of the day, with a volume ratio of 1.43. Thus, the evidence indicates that there is substantially greater volume following addition, and that the increase in volume is not spread equally throughout the day. Rather, it is concentrated at the beginning and end, producing a more pronounced U-shape for intraday volume. Furthermore, with the exception of period 4, all of the volume ratios are significantly

¹² We use the median because it is less sensitive to outliers. Inferences based on the mean, while more extreme, are similar.

smaller than period 5's ratio at the 1% level. In other words, the increase in volume at the end of the day is larger than the increase for other periods.

To control for possible changes in the number of shares outstanding over the test period, we repeat the tests for turnover. The results are very similar. The median daily turnover increases by 19%. Once again, this increase is most pronounced at the end of the day, where turnover increases by 36%. As with the volume ratios, period 5's turnover ratio is significantly larger than the ratios for the other four periods.

The next part of table 1 presents the number of trades before and after addition. Similar to the volume results, there is a large increase in the number of trades after index addition. For the day, the median number of trades increases by 36%. The increase at the end of the day is most striking, at 50%, and is significantly larger than the increase in the other periods. Finally, the volatility numbers presented at the end of table 1 provide some evidence both on the concentrated trading hypotheses and on information production. For reasons of robustness, we measure volatility as the mean absolute return.¹³ Volatility increases in every intraday period. The across-the-board rise in volatility is consistent with increased private information production. In other words, trading costs decline due to the increased volume of liquidity trade; this raises the returns to information production, and leads to more information being produced. (Later in this section, we present evidence that trading costs do, in fact, decline post-addition.)

The largest increase in volatility occurs at the end of the day. The large increases in volume and number of trades at the end of the day reflect the activity of index funds, and

the increase in volatility indicates that more information is being revealed at that time. This is consistent with the concentrated temporal equilibrium hypothesis: some of the informed trading has followed the increased liquidity trade to the end of the day, producing a new concentrated equilibrium.

As discussed above, the significant increase in volatility in every intraday period is consistent with increased information production after index addition. The desire of informed traders to hide their trades dictates that they should shift some of their trades to the end of the day, even if no additional private information is produced. Were this the case, however, the increased end-of-day volatility should have been accompanied by a reduction in volatility in other intraday intervals. We do not observe any such reductions in volatility.

As we noted earlier, this increased volatility is also consistent with price pressure. To distinguish between explanations based on information and price pressure, we carry out three tests. First, we examine the volatility of daily (close-close) returns. If mispricing is behind the high volatility in the five intraday periods, and is corrected by the end of the day, daily volatility will be unchanged. However, as shown in table 1, the increase in daily volatility, while smaller than the increase in volatility at the end of the day, is nevertheless highly significant in statistical terms. Second, for each stock, we examine the post-addition change in the correlation between the return at the end of the day and the returns during the following trading day. The cross-sectional median of the change in the correlation between the return in period 5 and the close-open return or the return in the

¹³ Measures of volatility based on the absolute return are more robust to outliers and to departures from normality than squared return-based measures. See Davidian and Carroll [1987] and Schwert [1989].

first half-hour on the following trading day is either positive or insignificantly different from zero. The median change in the correlation between the return in period 5 and the next day's close-close return is -0.03 and is significant only at the 12% level. This is driven entirely by the negative correlation between the period 5 return and the return in the second half-hour on that day. We are unable to construct a reversal story that explains why the reversal is confined to the second half-hour of the next trading day.

Finally, as a test of general intraday reversals, we also carry out a variance ratio test similar to that used by French and Roll [1986].¹⁴ Specifically, we divide the variance of the daily close-close returns, $Var(r_{c-c})$, by the sum of the variances of returns corresponding to the overnight period and the five intraday intervals:

$$VR_p = \frac{Var(r_{c-c,P})}{Var(r_{c-o,P}) + Var(r_{1,P}) + Var(r_{2,P}) + Var(r_{3,P}) + Var(r_{4,P}) + Var(r_{5,P})}$$

where P indicates the pre-addition or post-addition period. If the intraday returns are uncorrelated, VR_p will be one. Negative correlation will be reflected in a value of VR_p below one. Since, under the price pressure hypothesis, the buying of index funds will be manifested in eventual reversals, the post-addition change in the value of VR_p should be negative. The cross-sectional median of VR_p is 0.848 pre-addition and 0.850 after addition and the median paired change in VR_p is 0.006, insignificantly different from zero. Thus, while there is negative correlation (overreaction) both before and after index addition, intraday returns do not become more negatively correlated after addition.

¹⁴ The variance ratio test yields conclusions regarding the average pairwise correlation between returns corresponding to the five intraday periods and to the overnight period.

The conclusion that we draw from this supplementary analysis is that, while there may be some reversal of the price changes in the last period of the day (plausibly associated with the large increase in liquidity demand), this is, at best, incomplete. This evidence is, therefore, inconsistent with the prediction of the price pressure hypothesis.

The overall evidence for the trading environment indicates that it is significantly different after index addition. Volume increases dramatically, with significantly more of this increase coming at the end, rather than the middle, of the day. The raw number of trades also increases. Increased trading volume is consistent with enhanced index fund presence in the market after addition, particularly near the end of the day. Our next objective is to examine whether the increased trading by funds results in improved terms of trade. Here, we compare bid-ask spreads and the price impact of trades before and after index addition. First, however, we describe the formation of trade size categories.

Trade size category definition

We argue in Section 2 that index funds will execute large trades, and that informed traders have incentives to mimic their trading strategies. Accordingly, we examine the distribution of trade sizes before and after index addition. In order to classify trades as small, medium and large, we form quartiles of trade sizes corresponding to the pre-addition period. Quartile 1 represents the smallest 25% of the trades, quartile 2 the next largest trades, and so on. Due to the fact that trade sizes could vary through the day, these quartiles are created separately for the five intraday periods.

We carry these quartile breakpoints into the post-addition period. We then cumulate the percentage of transactions and trading volume in each of the four categories. This

procedure is chosen because forming trade size categories on the basis of shares traded and assigning the same trade size categories to all stocks is likely to obscure important differences across stocks. For instance, a thousand share transaction may be a large trade for a relatively thinly-traded stock and only a medium trade for another stock. In making their trading decisions, funds are likely to be sensitive to the prevailing distribution of transactions in the market. Given fund trading preferences, quartile 4 should capture the majority of fund trades.

Table 2 presents summary statistics for the cross-sectional distribution of quartile breakpoints. The table demonstrates that, while the means vary throughout the day, the three breakpoints are approximately 200, 500 and 1200 shares. More importantly, the standard deviations presented indicate that there is considerable variation in these breakpoints across firms, confirming the earlier supposition that what constitutes a “large” trade can be very different across stocks.

Spreads and the Price Impact of Trades

Since liquidity trading increases post-addition, traders should get better terms for trading. Thus, the bid-ask spread and the price impact of trades should decrease. We examine both in this section. Table 3 presents spread ratios calculated in the same manner as the volume ratios of the previous section. The cross-sectional median of the spread ratios is significantly less than one in every period, with approximately 60% of the stocks experiencing a reduction in the spread after addition. In the middle of the day, post-addition spreads are 98% of pre-addition levels.¹⁵ In the last hour of the day, they drop to

¹⁵ This results from cross-sectional averaging of the ratio of discrete spreads.

between 95% and 96% of pre-addition levels. Thus, it does seem that the increased liquidity trading results in smaller spreads, on average.

Traders are likely to be concerned with the price impact of their trades—perhaps more so than with the bid-ask spread.¹⁶ In order to study the price impact of trades for the stocks in our sample, we categorize each trade by its quartile. We also classify the trade as a buy or sell using the Lee-Ready [1991] algorithm. The price impact of a trade is the return computed from the quote midpoint in effect at the time of the trade (as long as it was recorded at least 5 seconds prior to the trade) to the first quote midpoint after the trade:

$$l_{s,q,k,t} = \log(mp_{s,q,+1} / mp_{s,q,-1})$$

where $l_{s,q,k,t}$ is the price impact for a trade in quartile q for stock s in intraday period k ($k=1, 2, \dots, 5$) on day t , and $mp_{s,q,+1}$ ($mp_{s,q,-1}$) is the quote midpoint succeeding (preceding) the trade.¹⁷ Then, for each stock, we compute the mean price impact by quartile. Using the pre-addition quartile breakpoints in the post-addition period, we follow the same procedure to obtain the mean post-addition price impact in each quartile, and thus the change in price impact from before to after addition.

Table 3 presents the medians of the mean price impacts before and after addition as well as the medians of their paired differences (post addition mean price impact–pre-addition mean price impact). For every intraday period, the median change in the price impact of a large buy order (quartile 4) is significantly negative. Thus, large buys move

¹⁶ The reasons include discreteness, quoted depth offsetting any spread effects, within-spread trades.

¹⁷ Our procedure is similar in principle to Hasbrouck [1991].

the quote midpoint less after addition than before addition. The results for large sell orders are similar, but not as consistently significant. When aggregated to the daily level, the change in the price impact of sales is significantly positive, however, indicating that, on average, large sell orders result in a smaller negative quote revision after addition than they did prior to it. These results are consistent with a general increase in the proportion of liquidity-motivated large trades. Specifically, because of index fund activity, large trades are less likely to be informationally-motivated in the post-addition period than in the pre-addition period.¹⁸

Several observations and implications follow from the results to this point. There clearly is increased liquidity trading post-addition. Trading increases substantially and most significantly at the end of the day. The most obvious source of this trading is index funds. Following index addition, several very large traders are now trading the stock for purely liquidity reasons. They mostly trade at the end of the day in order to minimize tracking error (although there may be some adjustments at the beginning of the next day after they have settled all of their inflows and outflows). All else being equal, the increased liquidity trading should result in better terms of trade execution—smaller spreads and smaller price impacts, especially for large trades.

An interesting prediction arises in equilibrium, however. As outlined in Section 2, informed traders have incentives to trade when liquidity traders are active (Admati and Pfleiderer [1988]). The general reduction in the price impact of trades affords them greater profits on their information. In addition, due to the fact that index funds make

¹⁸ Koski and Michaely [1998] present evidence that the market-maker adjusts prices and liquidity to a greater extent following trades that are more likely to be informationally-motivated.

large trades, informed traders are also able to make large trades (Barclay and Warner [1993]). Thus, these papers predict a concentrated equilibrium in time and trade size.

The results in table 3 support this implication. The post-addition price impact of large trades is lower in every period, but no period shows a significantly larger drop in the price impact than any other. Without a concentrated equilibrium of informed traders following liquidity traders (in choosing the time and size of their trades), one would expect the last half-hour of the day, with substantially greater liquidity trading, to be characterized by the largest reduction in price impacts. We suggest, however, that informed traders have moved into that half-hour, driving up the price impact until it is the same as in any other period and establishing a trading equilibrium. In the next section, we test for the presence of a concentrated equilibrium more directly using methods similar to those in Barclay and Warner [1993].

Informativeness of Trades

The results of the price impact tests in table 3 indicate that a large order results in a smaller concession after addition than before addition. This is consistent with a greater proportion of large trades being uninformed after addition. But this change in the price impact is an equilibrium outcome, reflecting the revised trading strategies of the informed traders, and the rational recognition of this revision by the market maker.

We are interested in how the strategy of informed traders changes in response to the altered distribution of liquidity trades. In order to infer the presence of informed traders, we use a measure of the *aggregate* information conveyed by trades, introduced by Barclay and Warner [1993]. This measure is based on the observation that permanent price

changes reflect new information (public or private) and temporary price changes are symptomatic of liquidity trades. Then, the aggregate price change in any interval will reflect new information revealed during the interval and the total price change associated with large trades will likewise reflect the aggregate information content of large trades. The fraction of the total price change occurring in a trade size category indicates the relative informativeness of that category.

This measure obviously depends on the price impact of each trade. However, it also depends on the distribution of trades. A reduction in the price impact per trade accompanied by a more substantial increase in the number of large trades will result in an increase in the aggregate informativeness of large trades, holding other price impacts and trades constant. Intuitively, a large increase in the number of uninformed trades that is not accompanied by an increase in informed trades will not change the aggregate amount of information discovery. The equilibrium reaction to such a purely uninformed trading increase would be an offsetting reduction in the price impact, leaving aggregate informativeness unchanged. To the extent that the additional trades do contain a non-trivial number of informed trades, the price impact does not drop as significantly, and the information measure registers an increase.

In order to implement the informativeness test, we proceed as follows. For each stock, we first calculate the price change for each trade as the deviation between the price at which this trade occurs and the price of the prior trade:

$$\text{Change in price} = Dp = \text{Price for Trade} - \text{Price for Previous Trade}.^{19}$$

¹⁹ As in Barclay and Warner [1993], we exclude the opening transaction because it is not a typical trade.

Next, we cumulate for each intraday interval and on each day, the total price change (preserving the sign) associated with trades in a particular trade category. Third, we divide this price change by the total price change for the particular interval on that day. For each stock, we then compute the mean of these ratios for the pre-addition and post-addition periods and the difference between the post and pre-addition means. Finally, we present the cross-sectional median of the stock statistics.

On the assumption that permanent price changes reflect private information, the proportions indicate the importance of trades of different sizes in providing information. If a particular trade size category contains more informed trading after index addition, the percentage of the total price move occurring in that category will increase after addition. It is possible that the cumulative price change for all trades increases after addition, but this measure will pick up the *relative* informativeness of trades of different sizes. Note that this measure is unlikely to be influenced by bid-ask bounce since the induced effects are equal but of opposite sign.²⁰

Table 4 presents the within-period distribution of the proportion of the price change occurring in each quartile of trades. Also reported are the changes in the proportion of each period's volume and transactions accounted for by each quartile (recall that the quartiles were computed and fixed based on the pre-addition period). As expected, most of the action is in the last half-hour. There is a pronounced shift in the period's volume

²⁰ In order to disentangle the effects of buys and sells, an alternative test uses the absolute value of the price change. The results of this test are not reported since they yield conclusions that are identical to those from the test based on the raw price change.

toward the largest quartile of trade sizes. Along with this shift, we see a decrease in the relative informativeness of trades in the smaller quartiles and an increase in the informativeness of trades in the larger quartiles. In fact, the table depicts a marked pattern of shifts in informativeness that matches the shifts in volume.

We directly test this conjecture by examining the correlation between the change in the proportion of the volume represented by a particular period-quartile with the change in the proportion of the price change for the same time and trade size category. Under the null hypothesis of no concentration in trade sizes and time, all of the changes in the distribution of trading volume are due only to the entrance of index funds, since informed traders do not change their strategies. Since the trading of funds is not informative, and since the location of informative trades is unchanged, the correlation is expected to be zero. However, the correlation between the change in the volume and price change proportions is 0.54, significant at the 1% level. This is further evidence in support of the concentrated equilibrium hypothesis. Put simply, index addition provides an exogenous shock to the size and timing of trades. This, in turn, leads to an endogenous response on the part of informed traders, who shift their trades to follow the changed pattern in liquidity trading.

Rejecting price pressure as an alternative explanation for our findings

An alternative explanation for the results in tables 3 and 4 is based on price pressure. This explanation runs as follows. The sample period has been characterized by an increase in the size of assets under indexing. As individual investors have directed their dollars at index funds, funds have, in turn, had to enter the market to buy stocks in the S&P 500.

The continual buying on the part of funds, exacerbated by the fact that funds make large trades, could leave the market maker short of inventory.

As a result, for the market maker's inventory position to be squared, a large number of small sell orders is required at other times in the day. (Funds will also place large buy orders at other times in the day, and the number of large sell orders is small, according to this story.) In the absence of an adequate number of such trades, the buying activity of index funds imposes inventory risk on the market maker. He responds by not reducing the price impact of large buys as much as he should have if these buys were part of an uncorrelated sequence of large trades. In addition, the increased frequency of large transactions in period 5 results in a greater proportion of price changes occurring in this category. Thus, price pressure provides a potential explanation for the results in tables 3 and 4 that does not involve the migration of informed trades.

For several reasons, we view this scenario as unlikely. Stocks in the S&P 500 have not experienced severe demand imbalances. From a theoretical standpoint, as well, the presence of large and recurring (on a daily basis) price premia for supplying liquidity should induce additional suppliers (other than the market maker) to enter the market. We have already presented evidence that intraday return correlations do not become more negative after addition. This indicates that reversals do not become more pronounced after index addition, and is contrary to the price pressure hypothesis. In addition, we have reported the fact that the bid-ask spread does not widen at the end of the day after

addition. This suggests that the market maker does not move the spread asymmetrically in response to order flow imbalances, in this case raising the ask price to deter buys.²¹

However, we conduct two further tests. First, we compare period 5 returns pre- and post-addition. If the market maker is worried about imbalances on the buy side, he should raise his price (the midpoint of the spread) in period 5. As a result, the return at the end of the day will increase after addition. For each stock, we calculate the mean return for period 5 before and after addition, as well as the paired difference. The cross-sectional median of the mean return is 0.000331 pre-addition and 0.000345 post-addition, both of which are significantly above zero, but the paired difference is not significant at conventional levels. Thus, consistent with previous evidence on common stocks (Harris [1986]), the end-of-day return is positive both before and after the index addition. However, it does not increase following addition, which is directly contrary to the predictions of the price pressure hypothesis.²²

Second, we also compute the fraction of the *daily* price change accounted for by large trades. If the price effect of large trades is transitory (and gets reversed within a day), the proportion of the daily price change occurring in the largest trade size category will not be significantly larger than before addition. We find that the change in the fraction of the daily price change occurring in the largest quartile of trades is 0.026 (with a p-value of 0.07), which is smaller than the change in the end-of-day proportion but still positive. Table 4 shows that no other intraday period sees an increase in either large trades or price

²¹ Brock and Kleidon [1992] model the effects of inelastic demand on the spread at the end of the day.

²² Once again, the means are skewed by outliers, and neither mean is significant.

proportions. The fact that large trades at the end of the day contribute to the daily price change is inconsistent with price pressure and instead indicates an informational role.

Taking the evidence from both of these tests into account, it does not appear to us as if price pressure drives our results.

5 Concluding comments

In this paper, we study the microstructure effects of index addition. Adding a stock to the S&P 500 index attracts permanent interest in that stock from index funds, which, we argue, fit the description of liquidity traders. Moreover, index funds have a strong preference for trading near the end of the day, and are likely to place large trades. Consequently, this event is characterized by an exogenous increase in the volume of liquidity trading, which is likely to be concentrated at particular times of the day and in particular trade sizes. These aspects of fund trading, in turn, change the optimal strategies of informed traders regarding the timing and size of their trades. Accordingly, index addition is ideal for the purposes of gaining insights into the behavior of informed traders.

Our key findings are as follows. Daily volume and intraday volume increase substantially after index addition. However, consistent with fund objectives (of minimizing tracking error), the largest increase in volume occurs in the last half-hour of the day. The increase in volume for other intraday intervals suggests that the increased post-addition volume has other sources as well (e.g. program trading), or that funds spread their trades across the day.

The volatility of daily and intraday returns increases after addition and, once again, the largest increase occurs at the end of the day. The increase in return volatility throughout

the day suggests that increased information production follows index addition. However, the large increase in end-of-day volatility also suggests that informed traders change the timing of their trades to coincide with the liquidity trades, as predicted by the model of Admati and Pfleiderer [1988].

The distribution of trade sizes shifts rightwards after addition, and the greatest shift occurs at the end of the day. This is consistent with fund activity, since funds are unlikely to make small trades. However, while the price impact of large trades declines, the decline is similar throughout the day. Hence it does not appear as if the market maker perceives a greater alleviation in the adverse selection problem due to large trades at the end of the day. This suggests that informed traders are placing large trades to a greater extent at the end of the day, since they are now able to ‘hide’ these trades more easily. This concentrated trading outcome is an equilibrium because the proportion of large trades coming from informed traders has not increased, owing to the large increase in the number of fund transactions.

In equilibrium, the trades of the informed reveal their private information. Consistent with the concentrated equilibrium, the aggregate information conveyed by large trades increases significantly after addition. Overall, our results suggest that informed traders alter their trading strategies to optimally match both the timing, and the size, of liquidity trades in the market.

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TABLE 1: TRADING ENVIRONMENT BEFORE AND AFTER INDEX ADDITION

For each stock in the sample (246), the mean turnover, volume, number of trades, and volatility (defined as mean absolute return) is computed for the pre-addition period (-260,-20) and the post-addition period (+20,+260). The medians of these statistics are presented. Ratios are computed on a stock-by-stock basis. For example, the daily turnover ratio is computed as the average daily turnover for stock i post-addition divided by the average for stock i pre-addition. The medians of these ratios are presented. Turnover, volume and trades corresponding to 10:30-3:00 are scaled by the number of half-hours.

		Interval					
		Daily	9:30-10:00	10:00-10:30	10:30-3:00	3:00-3:30	3:30-4:00
Turnover x 100	Before	0.223	0.021	0.012	0.013	0.008	0.013
	After	0.243	0.023	0.014	0.015	0.010	0.018
	<i>p-values for difference of medians</i>	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Turnover Ratios		1.138	1.096	1.128	1.100	1.136	1.241
<i>p-values for difference from 1</i>		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Volume	Before	95900	8800	5100	5600	3400	5900
	After	131700	12800	7800	7800	5800	10100
	<i>p-values for difference of medians</i>	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Volume Ratios		1.317	1.234	1.284	1.357	1.299	1.428
<i>p-values for difference from 1</i>		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Number of Trades	Before	59	6	5	4	4	6
	After	80	8	7	5	6	9
	<i>p-values for difference of medians</i>	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Nbr of Trades Ratios		1.354	1.326	1.314	1.366	1.368	1.432
<i>p-values for difference from 1</i>		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Volatility x 100	Before	0.978	0.313	0.079	0.347	0.275	0.273
	After	1.028	0.344	0.079	0.352	0.283	0.301
	<i>p-value for difference of medians</i>	(0.001)	(0.001)	(0.050)	(0.503)	(0.004)	(0.001)
Volatility Ratios		1.039	1.068	1.022	1.033	1.039	1.081
		(0.012)	(0.001)	(0.069)	(0.111)	(0.023)	(0.001)

TABLE 2: SUMMARY STATISTICS FOR TRADE SIZE QUARTILE BREAKPOINTS

This table presents the cross-sectional mean and standard deviation (in parentheses) of the breakpoints for the trade size quartiles for each intraday period. These breakpoints are formed using the pre-addition distribution of trades. The units are number of shares.

	Breakpoint 1 (Small)	Breakpoint 2 (Medium)	Breakpoint 3 (Large)
9:30-10:00	97.6 (66.2)	319.0 (181.7)	990.7 (1089.9)
10:00-10:30	189.5 (73.6)	509.3 (554.4)	1336.4 (1275.4)
10:30-3:00	178.1 (73.6)	466.8 (554.4)	1188.7 (1275.4)
3:00-3:30	183.8 (161.2)	474.5 (621.0)	1165.2 (1241.8)
3:30-4:00	191.1 (85.6)	520.6 (829.3)	1265.6 (1208.1)

TABLE 3: THE PRICE IMPACT OF TRADES

This table presents statistics on how index addition affected spreads and the price impact of trades. The price impact, defined as the revision in the quote midpoint associated with the trade, is the return computed from the midpoint immediately preceding the trade to the first midpoint after the trade. For each stock, the mean price impact for every trade size quartile in every intraday period is computed for the pre and post index addition periods. The cross-sectional medians of these numbers are presented. We also calculate the paired difference of these mean price impacts for each of the 246 stocks, and present the medians of these paired differences with the p-values for the null hypothesis that they are equal to zero. Finally, we compute the mean spread prevailing at the end of each intraday period for each stock for the two periods—before and after addition. The center column presents the medians of the distributions of paired spread ratios and p-values for the null hypothesis that they are equal to one. p-values are in parentheses.

	Sells				Buys				Spread Ratios
	Size Quartile				Size Quartile				
	Small	2	3	Large	Small	2	3	Large	
9:30-10:00									
Before	-0.0032	-0.0032	-0.0032	-0.0037	0.0027	0.0029	0.0034	0.0038	0.970 (0.056)
After	-0.0030	-0.0030	-0.0033	-0.0036	0.0029	0.0027	0.0030	0.0034	
Difference	0.0002 (0.2712)	0.0000 (0.3689)	0.0001 (0.4555)	0.0002 (0.1419)	0.0000 (0.3874)	-0.0004 (0.0002)	-0.0002 (0.0004)	-0.0004 (0.0002)	
10:00-10:30									
Before	-0.0019	-0.0023	-0.0026	-0.0029	0.0017	0.0022	0.0027	0.0029	0.984 (0.070)
After	-0.0019	-0.0022	-0.0025	-0.0028	0.0017	0.0021	0.0024	0.0028	
Difference	0.0001 (0.5531)	0.0001 (0.2067)	0.0001 (0.4893)	0.0001 (0.2097)	0.0000 (0.8767)	0.0000 (0.4848)	-0.0003 (0.0067)	-0.0001 (0.1078)	
10:30-3:00									
Before	-0.0018	-0.0021	-0.0024	-0.0026	0.0017	0.0020	0.0024	0.0027	0.982 (0.018)
After	-0.0017	-0.0021	-0.0023	-0.0026	0.0016	0.0020	0.0023	0.0027	
Difference	0.0000 (0.8784)	0.0001 (0.4075)	0.0001 (0.4243)	0.0002 (0.0568)	-0.0001 (0.3759)	-0.0001 (0.0715)	-0.0001 (0.0698)	-0.0002 (0.0254)	
3:00-3:30									
Before	-0.0017	-0.0018	-0.0023	-0.0025	0.0016	0.0018	0.0022	0.0026	0.957 (0.007)
After	-0.0016	-0.0019	-0.0022	-0.0024	0.0015	0.0020	0.0022	0.0025	
Difference	0.0000 (0.8252)	0.0000 (0.4989)	0.0001 (0.7165)	0.0002 (0.6804)	-0.0002 (0.0414)	0.0000 (0.6147)	0.0000 (0.3875)	-0.0002 (0.0415)	
3:30-4:00									
Before	-0.0015	-0.0019	-0.0022	-0.0024	0.0017	0.0020	0.0023	0.0026	0.966 (0.013)
After	-0.0016	-0.0019	-0.0021	-0.0024	0.0017	0.0020	0.0022	0.0025	
Difference	0.0001 (0.8925)	0.0001 (0.7466)	0.0002 (0.1584)	0.0001 (0.8037)	-0.0001 (0.2034)	0.0000 (0.4747)	-0.0001 (0.1945)	-0.0002 (0.0743)	
Daily									
Before	-0.0018	-0.0023	-0.0026	-0.0028	0.0018	0.0022	0.0026	0.0029	0.978 (0.005)
After	-0.0018	-0.0022	-0.0025	-0.0027	0.0018	0.0021	0.0024	0.0028	
Difference	0.0000 (0.4660)	0.0001 (0.2875)	0.0001 (0.2759)	0.0001 (0.0696)	-0.0001 (0.1704)	-0.0001 (0.0590)	-0.0002 (0.0111)	-0.0002 (0.0098)	

TABLE 4: CHANGES IN THE DISTRIBUTION OF VOLUME, TRADES AND PRICE CHANGES

This table presents statistics on the fraction of volume, trades and price changes in trade size quartiles in the five intraday periods and at the daily level. The median change from before to after addition to the S&P 500 index is presented for each intraday time period and each trade size quartile. For example, the median change in fraction of volume for Quartile 1 in the 9:30 to 10:00 period is computed as follows. The mean fraction of volume for each of the 246 stocks is computed for the pre-addition and post-addition periods. The difference of each of these paired means is then computed. The number presented is the median of the distribution of the 246 differences. p-values for the null hypothesis that the median equals zero are in parentheses.

		Size Quartile				Size Quartile				
		Small	2	3	Large		Small	2	3	Large
9:30-10:00	Volume	-0.0004 (0.001)	-0.0091 (0.001)	-0.0053 (0.013)	0.0214 (0.001)	Trades	0.0000 (0.007)	0.0000 (0.646)	0.0002 (0.307)	-0.0055 (0.413)
	Price Change	0.0000 (0.016)	0.0000 (0.928)	0.0074 (0.476)	-0.0073 (0.717)					
10:00-10:30	Volume	-0.0136 (0.001)	-0.0053 (0.012)	-0.0022 (0.062)	0.0193 (0.001)	Trades	-0.0168 (0.001)	0.0024 (0.166)	0.0012 (0.115)	-0.0020 (0.866)
	Price Change	-0.0151 (0.068)	-0.0151 (0.338)	0.0029 (0.500)	0.0092 (0.240)					
10:30-3:00	Volume	-0.0040 (0.001)	-0.0030 (0.004)	-0.0053 (0.010)	0.0156 (0.001)	Trades	-0.0155 (0.001)	-0.0006 (0.806)	0.0033 (0.030)	0.0001 (0.618)
	Price Change	-0.0075 (0.471)	-0.0035 (0.854)	0.0005 (0.949)	-0.0034 (0.529)					
3:00-3:30	Volume	-0.0227 (0.001)	-0.0051 (0.001)	-0.0033 (0.301)	0.0255 (0.001)	Trades	-0.0176 (0.001)	0.0039 (0.109)	0.0057 (0.028)	-0.0007 (0.705)
	Price Change	-0.0220 (0.010)	-0.0060 (0.328)	0.0058 (0.091)	0.0098 (0.393)					
3:30-4:00	Volume	-0.0232 (0.001)	-0.0141 (0.001)	-0.0106 (0.001)	0.0509 (0.001)	Trades	-0.0215 (0.001)	-0.0001 (0.574)	0.0048 (0.024)	0.0082 (0.001)
	Price Change	-0.0454 (0.001)	-0.0192 (0.048)	0.0048 (0.720)	0.0605 (0.001)					